

Labs for CMOS inverter ngspice simulations

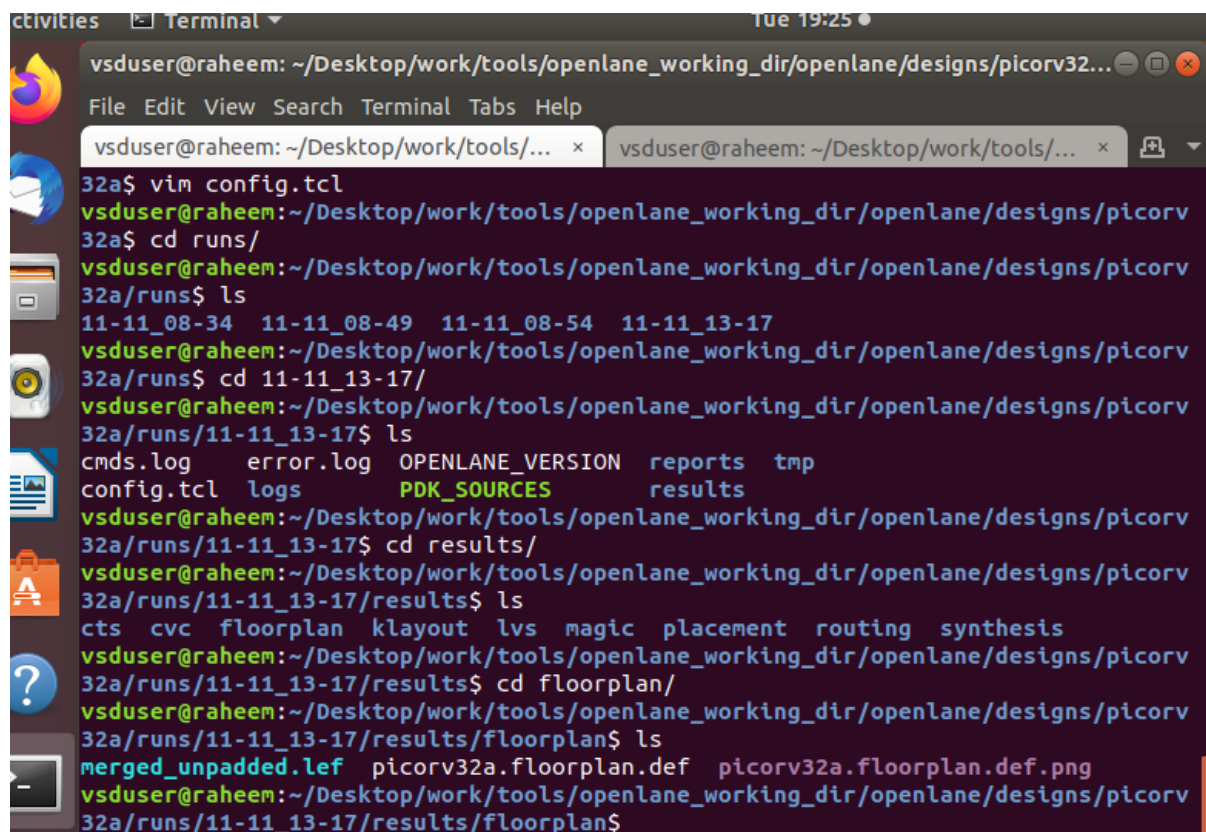
IO placer revision

This lab guides you through advanced flows in the OpenLane digital design automation environment, focusing on post-floorplan pin configuration and cell characterization. The core objective is to explore, modify, and verify input/output (IO) pin strategies using OpenLane's IO placer, then prepare the cell for post-layout simulation and characterization—including rise time, fall time, and propagation delay—for an inverter example. Steps entail navigating OpenLane run directories, inspecting and editing configuration variables, rerunning flows, and examining changes, followed by setup for detailed characterization.

Lab Steps

1. Inspect Completed Floorplan

- Ensure you are inside the latest OpenLane run directory (after synthesis and floorplanning).
- Open the results folder for the floorplan.

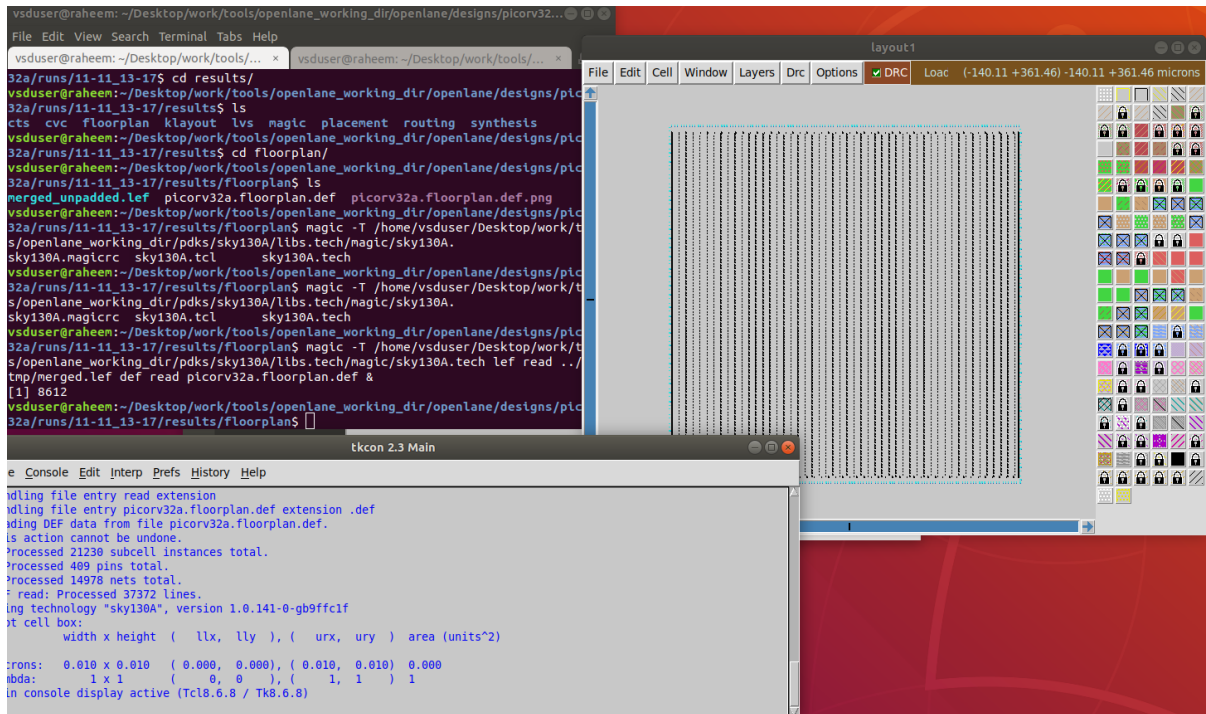


```
vsduser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32...
File Edit View Search Terminal Tabs Help
vsduser@raheem: ~/Desktop/work/tools/... x vsduser@raheem: ~/Desktop/work/tools/... x
32a$ vim config.tcl
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv
32a$ cd runs/
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv
32a/runs$ ls
11-11_08-34 11-11_08-49 11-11_08-54 11-11_13-17
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv
32a/runs$ cd 11-11_13-17/
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv
32a/runs/11-11_13-17$ ls
cmds.log error.log OPENLANE_VERSION reports tmp
config.tcl logs PDK_SOURCES results
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv
32a/runs/11-11_13-17$ cd results/
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv
32a/runs/11-11_13-17/results$ ls
cts cvc floorplan klayout lvs magic placement routing synthesis
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv
32a/runs/11-11_13-17/results$ cd floorplan/
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv
32a/runs/11-11_13-17/results/floorplan$ ls
merged_unpadded.lef picorv32a.floorplan.def picorv32a.floorplan.def.png
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv
32a/runs/11-11_13-17/results/floorplan$
```

- Locate and open the .def file to check the current pin configuration.

magic -T

```
/home/vsduser/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.tech/magic/sky130A.tech lef read ../../tmp/merged.lef def read picorv32a.floorplan.def &
```



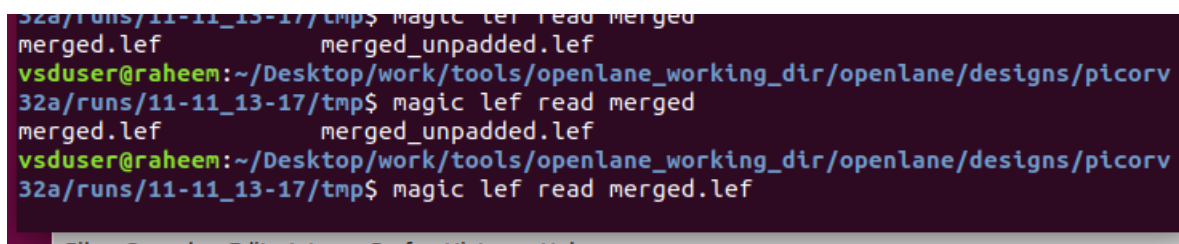
- Alternatively, view or copy the configuration from .def into a simpler editor (e.g., Notepad).

2. Read Current Pin Placement

- Use the LEF file for current pin data. If necessary, navigate up and into the temp directory to read the merged LEF.

```
cd ../../tmp
```

```
magic lef read merged.tab.lef
```



```

VERSION 5.7 ;
BUSBITCHARS "{}" ;
DIVIDERCHAR "/" ;

UNITS
  TIME NANoseconds 1 ;
  CAPACITANCE PICOfarADS 1 ;
  RESISTANCE OHMS 1 ;
  DATABASE MICRONS 1000 ;
END UNITS

MANUFACTURINGGRID 0.005 ;

PROPERTYDEFINITIONS
  LAYER LEF58 TYPE STRING ;
END PROPERTYDEFINITIONS

# High density, single height
SITE unithd
  SYMMETRY Y ;
  CLASS CORE ;
  SIZE 0.46 BY 2.72 ;
END unithd

# High density, double height
SITE unithdbl
  SYMMETRY Y ;
  CLASS CORE ;
  SIZE 0.46 BY 5.44 ;
END unithdbl

LAYER nwell
  TYPE MASTERSLICE ;
  PROPERTY LEF58_TYPE "TYPE N WELL ;" ;
END nwell

LAYER pwell
  TYPE MASTERSLICE ;
  PROPERTY LEF58_TYPE "TYPE P WELL ;" ;
END pwell

```

24,1

0%

- Check the pin arrangement visually in Magic. Pins are equally spaced after the initial floorplan and visible using left/right mouse actions.

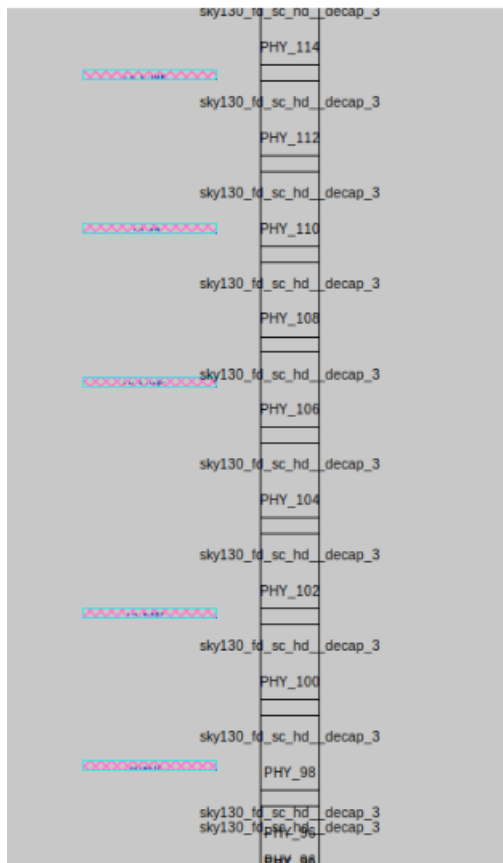
- Alternatively, view or copy the configuration from .def into a simpler editor (e.g., Notepad).

2. Read Current Pin Placement

- Use the LEF file for current pin data. If necessary, navigate up and into the temp directory to read the merged LEF.

cd ../../temp

magic lef read merged.tab.lef



3. Change IO Pin Placement Strategy

Enter the OpenLane configuration folder:

```
cd <openlane_project>/configuration
```

Identify the IO placer strategy variable (usually FP_IO_MODEL) within the relevant floorplan configuration file.

```
gvim floorplan.tcl
```

```
vsduser@raheem: ~/Desktop/work/tools/openl... x vsduser@raheem: ~/Desktop/work/tools/openl... x
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.

# Floorplan defaults
set ::env(FP_IO_VMETAL) 3
set ::env(FP_IO_HMETAL) 4

set ::env(FP_SIZING) relative
set ::env(FP_CORE_UTIL) 50
set ::env(FP_CORE_MARGIN) 0
set ::env(FP_ASPECT_RATIO) 1

set ::env(FP_PDN_VOFFSET) 16.32
set ::env(FP_PDN_VPITCH) 153.6
set ::env(FP_PDN_HOFFSET) 16.65
set ::env(FP_PDN_HPITCH) 153.18

set ::env(FP_PDN_AUTO_ADJUST) 1

set ::env(FP_PDN_CORE_RING) 0
set ::env(FP_PDN_ENABLE_RAILS) 1

set ::env(FP_PDN_CHECK_NODES) 1

set ::env(FP_IO_MODE) 1; # 0 matching mode - 1 random equidistant mode
set ::env(FP_IO_HLENGTH) 4
set ::env(FP_IO_VLENGTH) 4
set ::env(FP_IO_VEXTEND) -1
set ::env(FP_IO_HEXTEND) -1
set ::env(FP_IO_VTHICKNESS_MULT) 2
set ::env(FP_IO_HTHICKNESS_MULT) 2
```

- Note the current mode (e.g., FP_IO_MODEL = 1 for equidistant pins).
- Change the strategy by setting the variable to another supported mode (e.g., 2 for the Hungarian algorithm):

```
set ::env(FP_IO_MODE) 2
```

```
File Edit View Search Terminal Tabs Help
vsduser@raheem: ~/Desktop/work/tools/openl... x vsduser@raheem: ~/Desktop/work/tools/openl... x
# distributed under the license is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.

# Floorplan defaults
set ::env(FP_IO_VMETAL) 3
set ::env(FP_IO_HMETAL) 4

set ::env(FP_SIZING) relative
set ::env(FP_CORE_UTIL) 50
set ::env(FP_CORE_MARGIN) 0
set ::env(FP_ASPECT_RATIO) 1

set ::env(FP_PDN_VOFFSET) 16.32
set ::env(FP_PDN_VPITCH) 153.6
set ::env(FP_PDN_HOFFSET) 16.65
set ::env(FP_PDN_HPITCH) 153.18

set ::env(FP_PDN_AUTO_ADJUST) 1

set ::env(FP_PDN_CORE_RING) 0
set ::env(FP_PDN_ENABLE_RAILS) 1

set ::env(FP_PDN_CHECK_NODES) 1

set ::env(FP_IO_MODE) 2; # 0 matching mode - 1 random equidistant mode
set ::env(FP_IO_HLENGTH) 4
set ::env(FP_IO_VLENGTH) 4
set ::env(FP_IO_VEXTEND) -1
set ::env(FP_IO_HEXTEND) -1
set ::env(FP_IO_VTHICKNESS_MULT) 2
set ::env(FP_IO_HTHICKNESS_MULT) 2
```

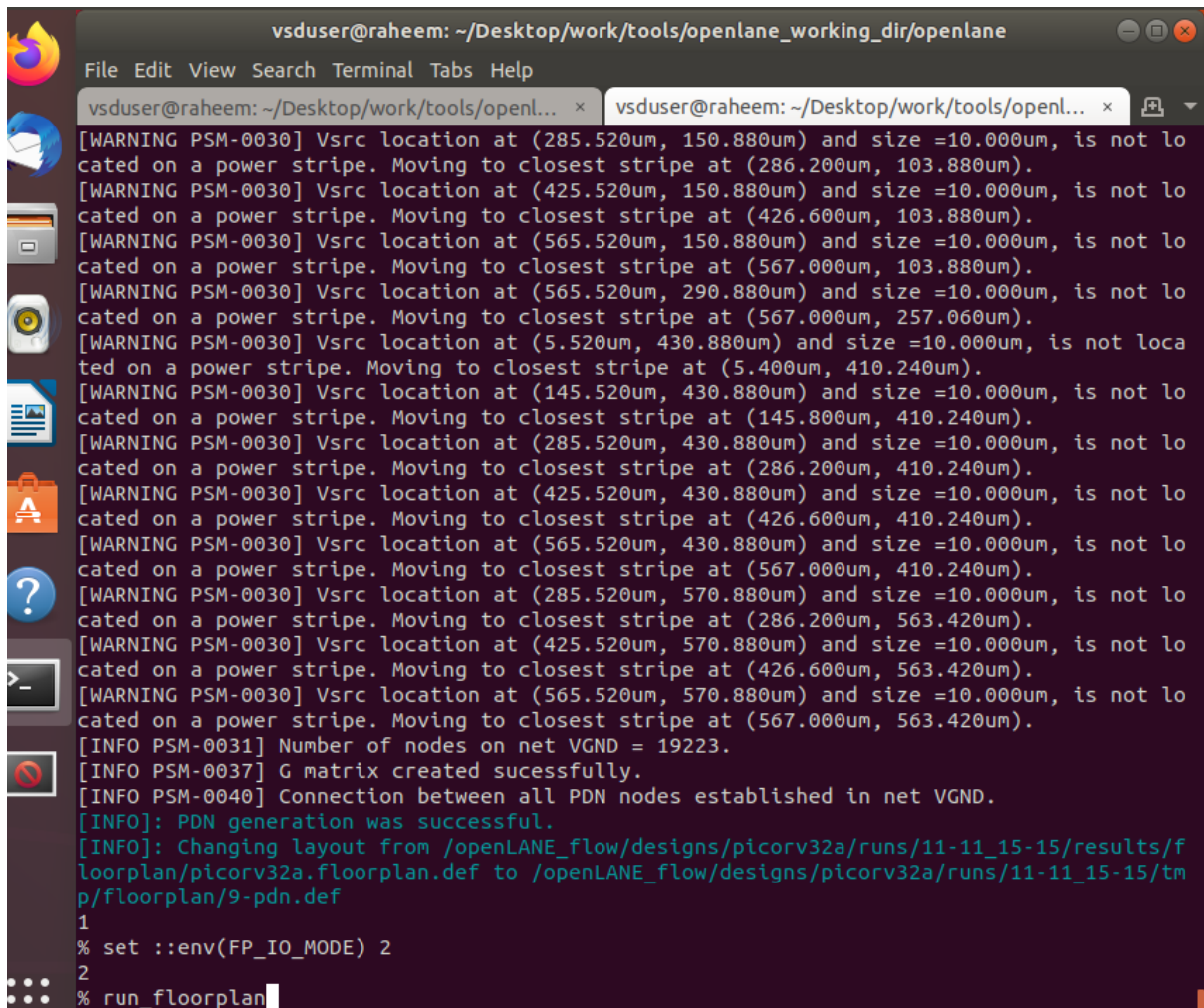
- Note the current mode (e.g., FP_IO_MODEL = 1 for equidistant pins).
- Change the strategy by setting the variable to another supported mode (e.g., 2 for the Hungarian algorithm):

```
set ::env(FP_IO_MODE) 2
```

Rerun Floorplan with New Settings

- Run the floorplanning process again using the updated IO placer setting:

```
run_floorplan
```



A terminal window titled "vdsuser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane". The window has a menu bar with "File", "Edit", "View", "Search", "Terminal", "Tabs", and "Help". Below the menu bar, there are two tabs, both showing the same path. The terminal output consists of several warning messages from the PSM (Power Signal Manager) component, followed by information messages and a successful PDN generation status.

```
vdsuser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane
File Edit View Search Terminal Tabs Help
vdsuser@raheem: ~/Desktop/work/tools/openlane... x vdsuser@raheem: ~/Desktop/work/tools/openlane... x
[WARNING PSM-0030] Vsrc location at (285.520um, 150.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (286.200um, 103.880um).
[WARNING PSM-0030] Vsrc location at (425.520um, 150.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (426.600um, 103.880um).
[WARNING PSM-0030] Vsrc location at (565.520um, 150.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (567.000um, 103.880um).
[WARNING PSM-0030] Vsrc location at (565.520um, 290.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (567.000um, 257.060um).
[WARNING PSM-0030] Vsrc location at (5.520um, 430.880um) and size =10.000um, is not loca
ted on a power stripe. Moving to closest stripe at (5.400um, 410.240um).
[WARNING PSM-0030] Vsrc location at (145.520um, 430.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (145.800um, 410.240um).
[WARNING PSM-0030] Vsrc location at (285.520um, 430.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (286.200um, 410.240um).
[WARNING PSM-0030] Vsrc location at (425.520um, 430.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (426.600um, 410.240um).
[WARNING PSM-0030] Vsrc location at (565.520um, 430.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (567.000um, 410.240um).
[WARNING PSM-0030] Vsrc location at (285.520um, 570.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (286.200um, 563.420um).
[WARNING PSM-0030] Vsrc location at (425.520um, 570.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (426.600um, 563.420um).
[WARNING PSM-0030] Vsrc location at (565.520um, 570.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (567.000um, 563.420um).
[INFO PSM-0031] Number of nodes on net VGND = 19223.
[INFO PSM-0037] G matrix created sucessfully.
[INFO PSM-0040] Connection between all PDN nodes established in net VGND.
[INFO]: PDN generation was successful.
[INFO]: Changing layout from /openLANE_flow/designs/picorv32a/runs/11-11_15-15/results/f
loorplan/picorv32a.floorplan.def to /openLANE_flow/designs/picorv32a/runs/11-11_15-15/tm
p/floorplan/9-pdn.def
1
% set ::env(FP_IO_MODE) 2
2
% run_floorplan
```



```
Activities Terminal Tue 20:50
vdsuser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane
File Edit View Search Terminal Tabs Help
vdsuser@raheem: ~/Desktop/work/tools/openl... x vdsuser@raheem: ~/Desktop/work/tools/openl... x
[WARNING PSM-0030] Vsrc location at (145.520um, 10.880um) and size =10.000um, is not located on a power stripe. Moving to closest stripe at (145.800um, 107.180um).
[WARNING PSM-0030] Vsrc location at (285.520um, 10.880um) and size =10.000um, is not located on a power stripe. Moving to closest stripe at (286.200um, 107.180um).
[WARNING PSM-0030] Vsrc location at (425.520um, 10.880um) and size =10.000um, is not located on a power stripe. Moving to closest stripe at (426.600um, 107.180um).
[WARNING PSM-0030] Vsrc location at (565.520um, 10.880um) and size =10.000um, is not located on a power stripe. Moving to closest stripe at (567.000um, 107.180um).
[WARNING PSM-0030] Vsrc location at (285.520um, 150.880um) and size =10.000um, is not located on a power stripe. Moving to closest stripe at (286.200um, 107.180um).
[WARNING PSM-0030] Vsrc location at (425.520um, 150.880um) and size =10.000um, is not located on a power stripe. Moving to closest stripe at (426.600um, 107.180um).
[WARNING PSM-0030] Vsrc location at (565.520um, 150.880um) and size =10.000um, is not located on a power stripe. Moving to closest stripe at (567.000um, 107.180um).
[WARNING PSM-0030] Vsrc location at (565.520um, 290.880um) and size =10.000um, is not located on a power stripe. Moving to closest stripe at (567.000um, 260.360um).
[WARNING PSM-0030] Vsrc location at (5.520um, 430.880um) and size =10.000um, is not located on a power stripe. Moving to closest stripe at (5.400um, 413.540um).
[WARNING PSM-0030] Vsrc location at (145.520um, 430.880um) and size =10.000um, is not located on a power stripe. Moving to closest stripe at (145.800um, 413.540um).
[WARNING PSM-0030] Vsrc location at (285.520um, 430.880um) and size =10.000um, is not located on a power stripe. Moving to closest stripe at (286.200um, 413.540um).
[WARNING PSM-0030] Vsrc location at (425.520um, 430.880um) and size =10.000um, is not located on a power stripe. Moving to closest stripe at (426.600um, 413.540um).
[WARNING PSM-0030] Vsrc location at (565.520um, 430.880um) and size =10.000um, is not located on a power stripe. Moving to closest stripe at (567.000um, 413.540um).
[INFO PSM-0031] Number of nodes on net VGND = 19223.
[INFO PSM-0037] G matrix created successfully.
[INFO PSM-0040] Connection between all PDN nodes established in net VGND.
[INFO]: PDN generation was successful.
[INFO]: Changing layout from /openLANE_flow/designs/picorv32a/runs/11-11_15-15/results/floorplan/picorv32a.floorplan.def to /openLANE_flow/designs/picorv32a/runs/11-11_15-15/tmp/floorplan/14-pdn.def
1
```



```
File Edit View Search Terminal Tabs Help
vdsuser@raheem: ~/Desktop/work/tools/openl... x vdsuser@raheem: ~/Desktop/work/tools/openl... x
[WARNING PSM-0030] Vsrc location at (285.520um, 150.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (286.200um, 103.880um).
[WARNING PSM-0030] Vsrc location at (425.520um, 150.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (426.600um, 103.880um).
[WARNING PSM-0030] Vsrc location at (565.520um, 150.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (567.000um, 103.880um).
[WARNING PSM-0030] Vsrc location at (565.520um, 290.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (567.000um, 257.060um).
[WARNING PSM-0030] Vsrc location at (5.520um, 430.880um) and size =10.000um, is not loca
ted on a power stripe. Moving to closest stripe at (5.400um, 410.240um).
[WARNING PSM-0030] Vsrc location at (145.520um, 430.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (145.800um, 410.240um).
[WARNING PSM-0030] Vsrc location at (285.520um, 430.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (286.200um, 410.240um).
[WARNING PSM-0030] Vsrc location at (425.520um, 430.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (426.600um, 410.240um).
[WARNING PSM-0030] Vsrc location at (565.520um, 430.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (567.000um, 410.240um).
[WARNING PSM-0030] Vsrc location at (285.520um, 570.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (286.200um, 563.420um).
[WARNING PSM-0030] Vsrc location at (425.520um, 570.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (426.600um, 563.420um).
[WARNING PSM-0030] Vsrc location at (565.520um, 570.880um) and size =10.000um, is not lo
cated on a power stripe. Moving to closest stripe at (567.000um, 563.420um).
[INFO PSM-0031] Number of nodes on net VGND = 19223.
[INFO PSM-0037] G matrix created successfully.
[INFO PSM-0040] Connection between all PDN nodes established in net VGND.
[INFO]: PDN generation was successful.
[INFO]: Changing layout from /openLANE_flow/designs/picorv32a/runs/11-11_15-15/results/f
loorplan/picorv32a.floorplan.def to /openLANE_flow/designs/picorv32a/runs/11-11_15-15/tm
p/floorplan/9-pdn.def
1
% set ::env(FP_IO_MODE) 2
2
%
```

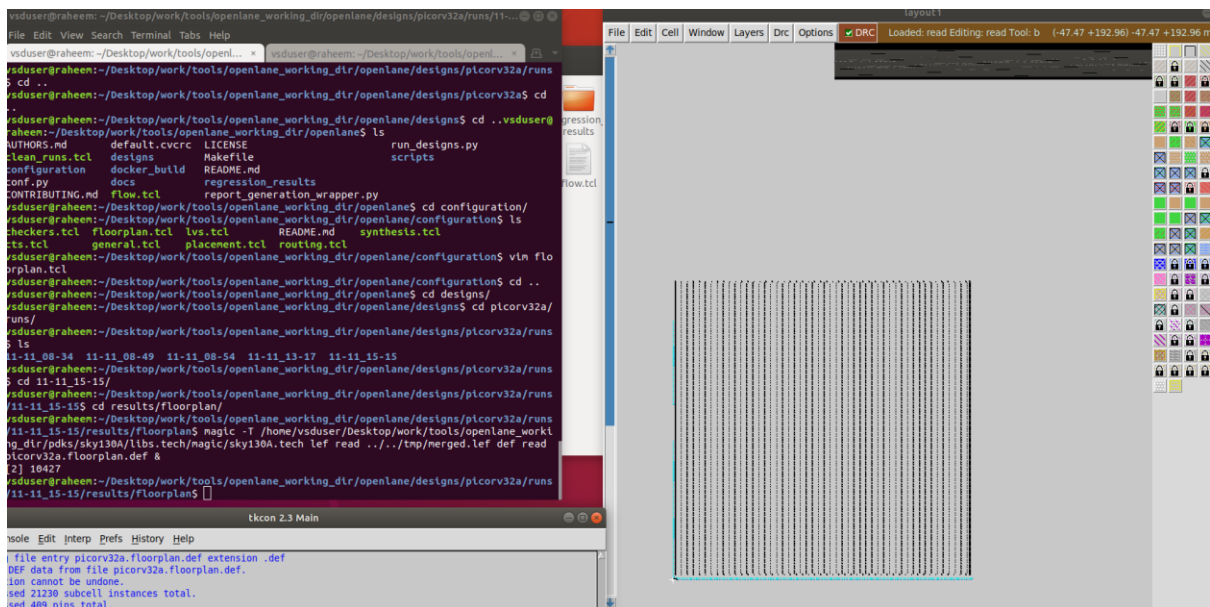
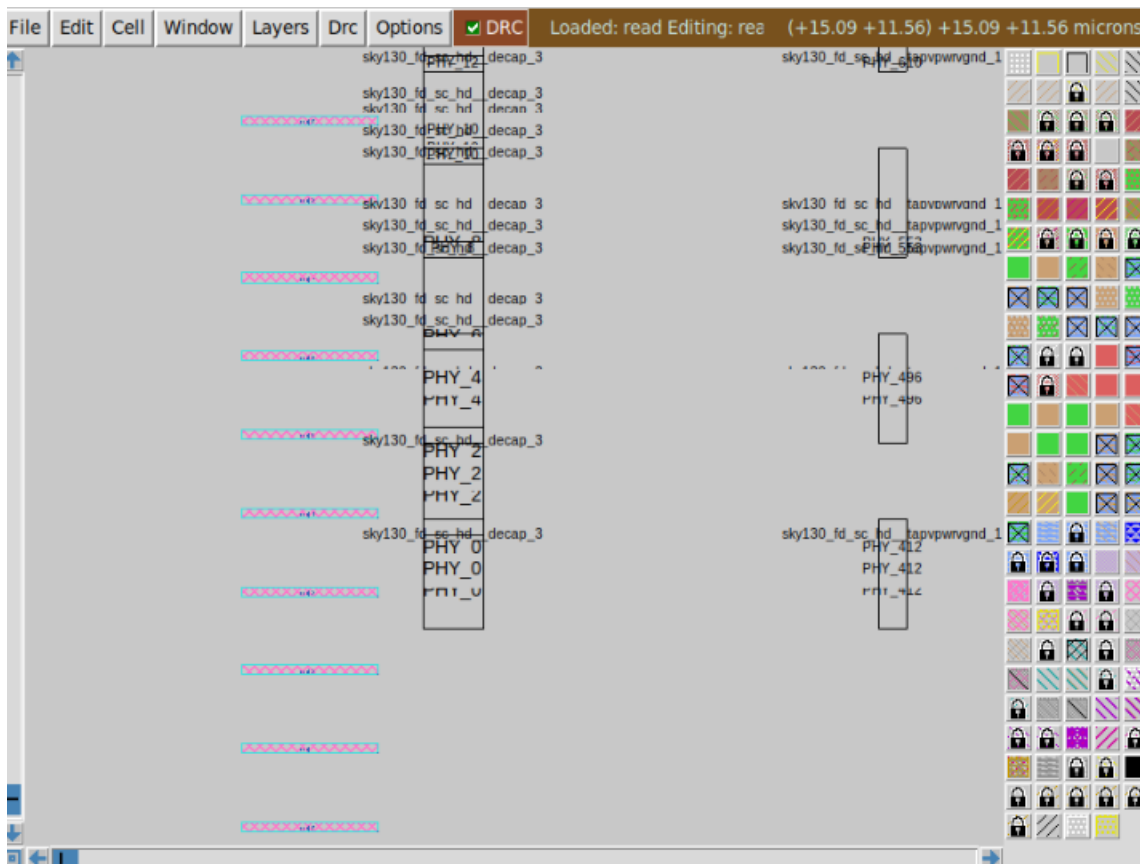
Verify Updated Pin Configuration

- Return to the results folder for the floorplan.

```
cd <openlane_project>/<latest_run>/results/floorplan
```

```
magic -T
```

```
/home/vdsuser/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.tech/mag
ic/sky130A.tech lef read ../../tmp/merged.lef def read picorv32a.floorplan.def &
```



- Inspect the new pin configuration in Magic; pins should no longer be equidistant—now stacked per the selected algorithm (e.g., Hungarian).

6. Reset and Iterate as Needed

- Any time you need to update configuration:
 - Reset the variable value.
 - Rerun the corresponding OpenLane flow.

7. Preparing for Cell Characterization

- The next phase is focused on **cell characterization** (inverter as the example), to measure parameters:
 - Rise time (t_{rise})
 - Fall time (t_{fall})
 - Propagation delay (t_{pd})
 - Rise propagation delay (t_{pr})
 - Fall propagation delay (t_{pf})

(Equations and detailed characterization will be conducted in following steps or videos. No commands in this step from the transcript.)

8. Setup for Parallel Work

- Open a parallel terminal for characterization work in the OpenLane root directory.

```
cd <openlane_project>
```

SPICE deck creation for CMOS inverter

This documentation explains the theoretical foundation behind voltage transfer characteristics in CMOS circuits and the foundational concepts of SPICE simulation as applied to CMOS inverters. It covers the symbolism and connectivity requirements for PMOS and NMOS devices, the reasoning behind component sizing, the significance of the load capacitor, and the definition and necessity of nodes within circuit netlists used by SPICE simulators.

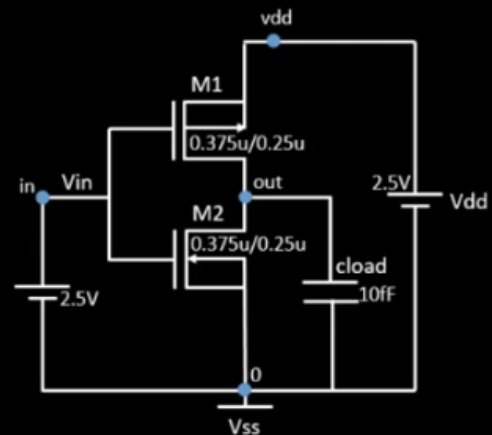
Core Concepts

Voltage Transfer Characteristics of CMOS

In CMOS circuits, the *voltage transfer characteristic* describes how the output voltage responds as the input voltage changes. This relationship is key to understanding how CMOS inverters switch states and operate in digital logic.

SPICE deck

- Component connectivity
- Component values
- Identify 'nodes'
- Name 'nodes'



SPICE Simulation Overview

SPICE (Simulation Program with Integrated Circuit Emphasis) simulators use mathematical models to approximate the electrical behavior of actual devices. Each component in a simulation (such as PMOS, NMOS, resistors, or capacitors) is defined by a mathematical model that captures its essential electrical characteristics. Simulators leverage these models to compute expected circuit behavior under specified conditions.

Netlist and Connectivity

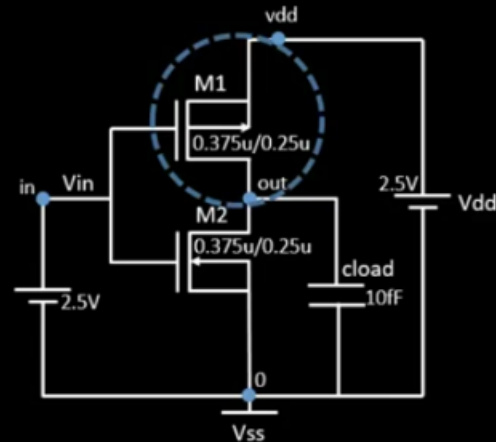
A SPICE simulation is based on a *netlist*—a textual description detailing all circuit components, their interconnections, and the configuration of the simulation (e.g., stimulus, output points). The netlist must communicate:

- The types and models of each device (e.g., PMOS, NMOS)
- The specific circuit instantiation parameters (width, length, etc.)
- All nodes (junctions where two or more components connect)
- Input signals, supply voltages, and outputs

The connectivity of the *substrate* terminal for MOSFETs must be explicitly defined, as it influences electrical characteristics like the threshold voltage.

SPICE deck

```
*** MODEL Descriptions ***  
*** NETLIST Description ***  
M1 out in vdd vdd pmos W=0.375u L=0.25u  
M2 out in 0 0 nmos W=0.375u L=0.25u
```



Nodes and Node Naming

A *node* is a point where device terminals connect—each component is defined between two or more named nodes. In SPICE, proper node identification is essential, as components are referenced in terms of their connecting nodes (e.g., IN, OUT, VDD, 0).

For example, the output load capacitor would be defined as connected between the nodes OUT and 0 (ground).

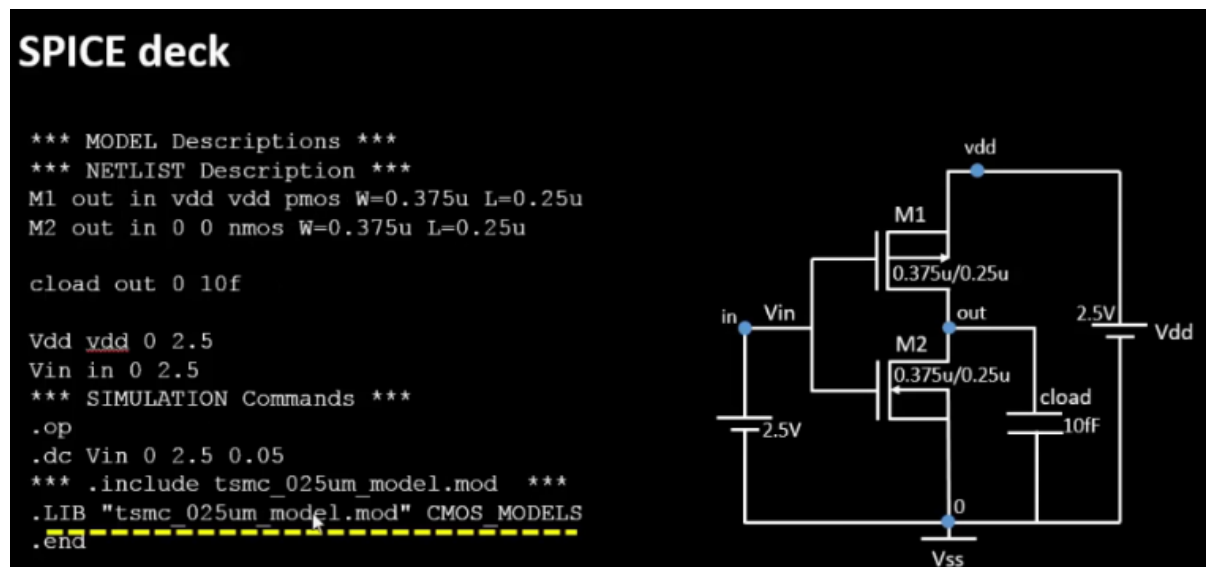
Key Points

- **SPICE Simulators** simulate electrical circuits using mathematical models attached to each component. These models approximate real-world device behavior.
- **Netlist construction** is the essential first step in simulation, encoding every component, connection, node, and necessary parameter.
- **PMOS and NMOS symbols** differ in arrow direction—the arrow out for PMOS, in for NMOS—encoding device type and current direction.
- **Substrate connection** must be specified for every MOSFET, as the substrate (or body) determines threshold voltage and can affect device behavior.
- **Component sizing** (i.e., width and length of MOSFETs) determines electrical strength. Typically, PMOS is sized wider than NMOS to balance drive strengths due to mobility differences.
- **Load capacitance** is fundamental in digital circuits, representing the output node's total capacitance. While sometimes assumed for simplicity (e.g., 10 fF), it typically requires calculation from circuit topology and is crucial for timing analysis.

- **Voltage scaling** is tied to technology size; smaller geometries use lower supply voltages, commonly scaling V_{DD} with channel length.
- **Nodes** define the connectivity in SPICE models. Each part must clearly indicate which nodes (IN, OUT, VDD, 0, etc.) it connects between. Consistent naming ensures correct simulation behavior.

SPICE simulation lab for CMOS inverter

This documentation explains the theoretical principles and connectivity behind the SPICE deck for a **CMOS inverter**. It covers the structure and roles of PMOS and NMOS transistors within the inverter, the connection of typical components such as load capacitors and voltage sources, the concept and purpose of voltage transfer characteristics, and the significance of transistor sizing (width and length) in determining inverter behavior.



Core Concepts

CMOS Inverter Structure

A CMOS inverter consists of a PMOS transistor and an NMOS transistor, connected in series between the supply voltage (V_{DD}) and ground. The output is taken from the common connection point between the PMOS drain and NMOS drain. The gate terminals of both transistors are tied together as the input. The substrate (body) connections for PMOS and NMOS are typically connected to V_{DD} and ground, respectively.

Signal Nodes and Connectivity

- **PMOS Connections:**

- Drain: Connected to the output node
- Gate: Connected to input node
- Source/Substrate: Connected to V_{DD}
- **NMOS Connections:**
 - Drain: Connected to output node
 - Gate: Connected to input node
 - Source/Substrate: Connected to ground (node zero)

This specific arrangement produces inverting logic behavior: high input yields low output, and vice versa.

Load Capacitance and Input/Output Voltages

The output node is typically loaded with a capacitor to simulate real circuit loads. The load capacitor (C_{load}) is connected between the output and ground and influences the dynamic response (slew rate, delay) of the inverter. The input and supply voltages are specified to establish operating conditions, commonly tying them to reference values, such as 2.5V.

Voltage Transfer Characteristics (VTC)

The **voltage transfer characteristics** plot the output voltage (V_{out}) against the swept input voltage (V_{in}). This curve reveals the inverter's switching threshold and defines its digital transition region. To generate VTC:

- Sweep V_{in} from 0 V to V_{DD} (e.g., 2.5 V), in defined steps (e.g., 0.05 V).
- Measure V_{out} for each V_{in} .

This process is essential for analyzing inverter behavior, noise margins, and switching performance.

SPICE Model Files

SPICE model files encapsulate the parameters governing transistor behavior. For each device type (NMOS and PMOS), the model includes critical process-dependent attributes such as:

- Oxide thickness (t_{ox})
- Threshold voltage (V_{th})
- Channel dimensions (Width W , Length L)
- Mobility and process coefficients (K_1 , K_2 , K_3 , etc.)

These parameters reflect the fabrication technology (e.g., 250nm process), and are referenced by model names such as 'NMOS' and 'PMOS'.

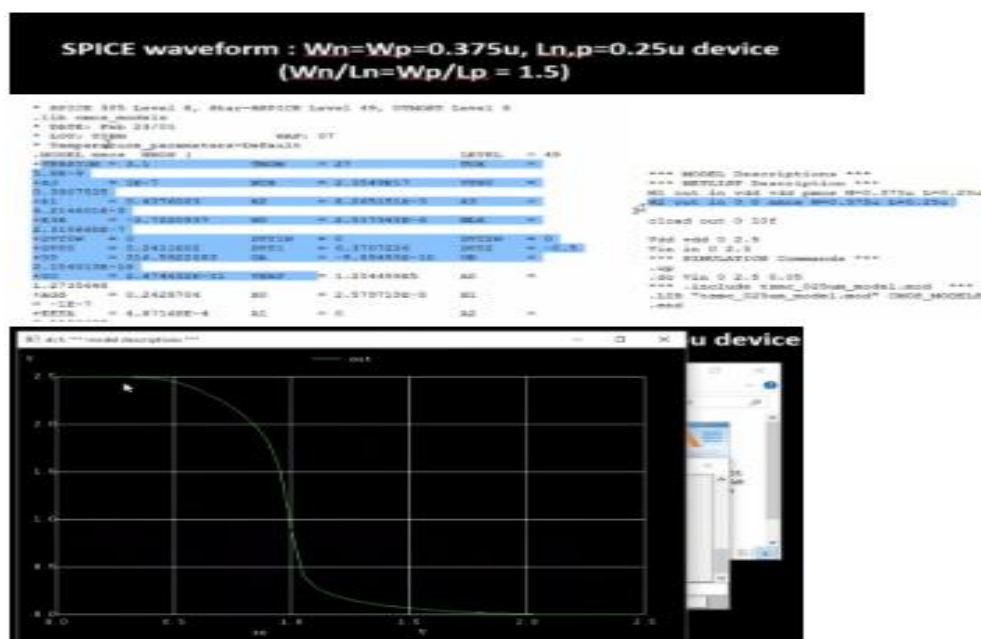
Sizing and W/L Ratio

The inverter's static and dynamic behavior is strongly affected by the dimensions of the PMOS and NMOS channels:

- W : Channel width
- L : Channel length
- W/L : Ratio influencing drive strength

For balanced operation, designers may set $W_{PMOS}/L_{PMOS} = W_{NMOS}/L_{NMOS}$, but commonly W_{PMOS} is increased relative to W_{NMOS} (e.g., $2.5\times$) to account for lower carrier mobility in PMOS, shifting the switching point and improving symmetric voltage transfer.

Example calculation: $W_{PMOS} = 2.5 \times W_{NMOS}$



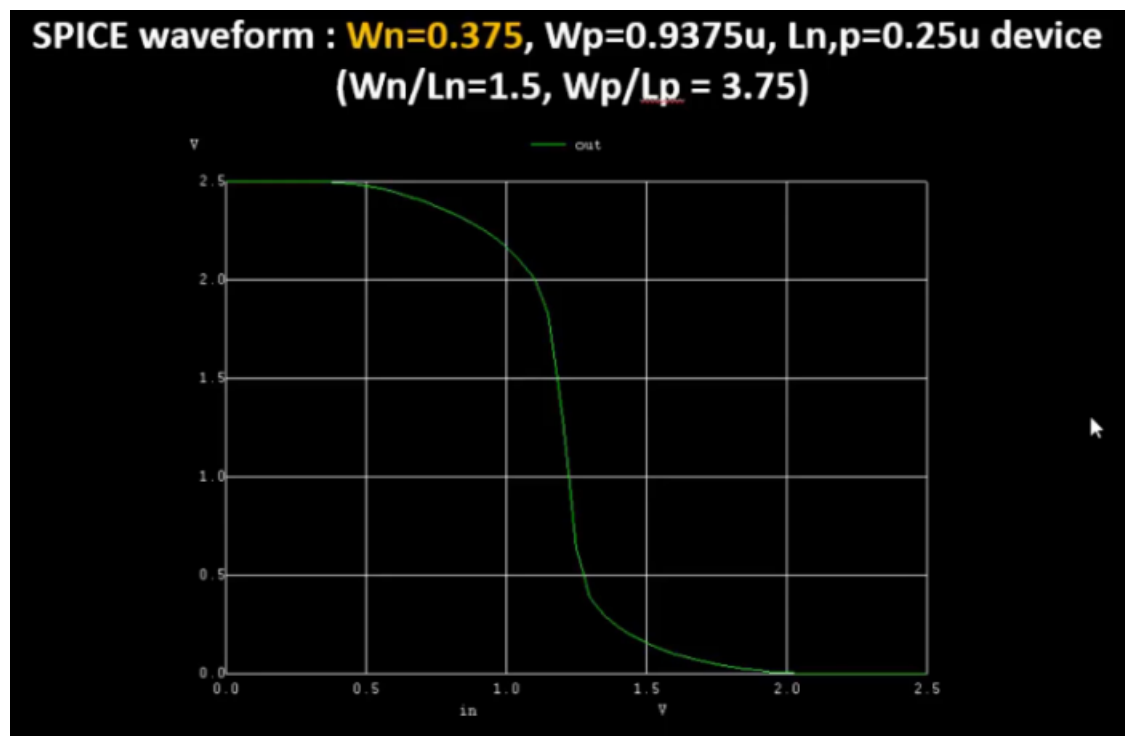
Key Points

- **CMOS inverter operation relies on complementary switching of PMOS and NMOS:** high input turns off PMOS and turns on NMOS, pulling output low; low input turns on PMOS and turns off NMOS, pulling output high.
- **Transistor connectivity defines logic inversion:** drain connected to output, gates to input, sources/substrates to supply and ground.
- **Load capacitance accurately reflects output dynamics and is essential for transfer characteristic analysis.**

- **SPICE simulation involves sweeping V_{in} to observe V_{out} :** V_{in} is incremented in steps, output voltage is recorded, and VTC plotted.
- **Transistor sizing (W/L ratio) is tuned for optimal inverter switching performance:** A wider PMOS typically shifts VTC for improved noise margins and balanced output.
- **SPICE model files embed all key process parameters, defining detailed device behavior for accurate simulation.**

Switching Threshold V_m

This documentation focuses on the theoretical analysis of SPICE simulations for CMOS inverters, particularly examining the impact of transistor sizing (PMOS vs NMOS) and the concept of switching threshold. Two key scenarios are considered: one where PMOS and NMOS have equal width-to-length (W/L) ratios, and another where PMOS is sized 2.5 times larger than NMOS. The analysis demonstrates the robustness of CMOS logic and explores how device sizing affects inverter characteristics such as switching threshold and current flow.

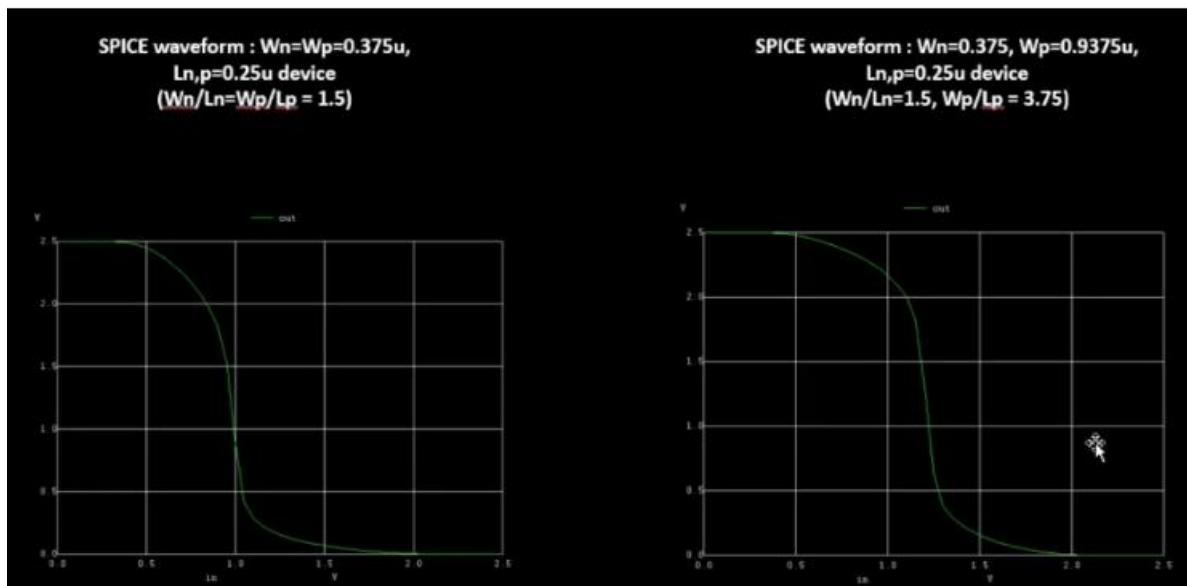


Core Concepts

CMOS Inverter Sizing

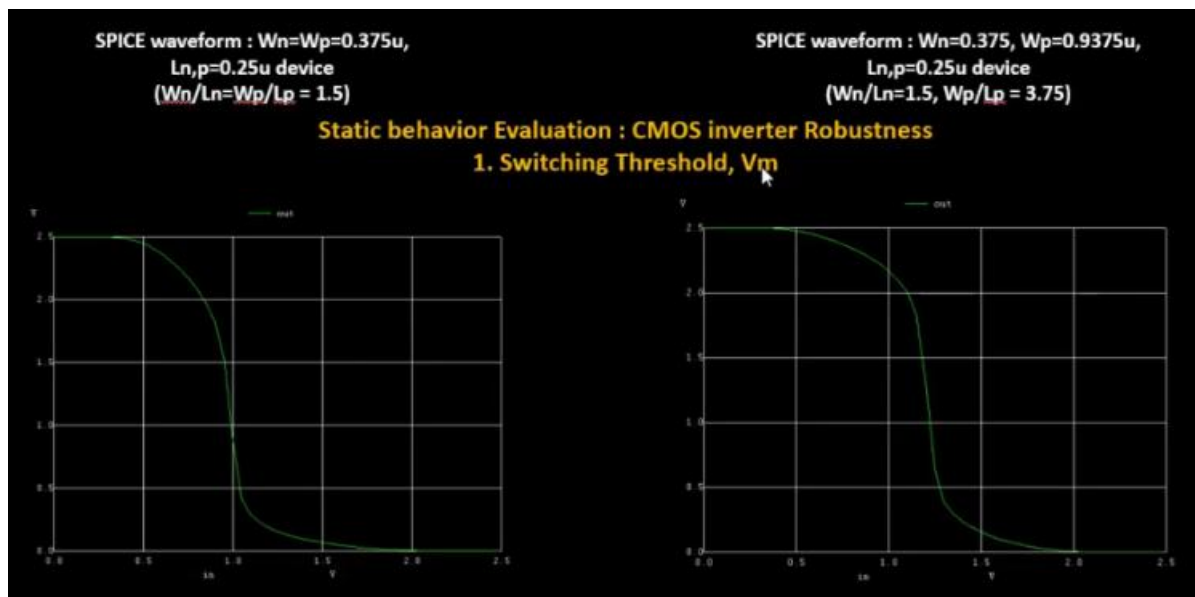
- **Device Sizing Scenarios:**

- **Equal W/L Ratios:** PMOS and NMOS have identical width and length, resulting in identical W/L ratios.
- **Larger PMOS:** PMOS width is increased to be 2.5 times the NMOS width, while lengths remain equal. This changes the PMOS W/L ratio relative to NMOS.
- **Calculation of W/L Ratios:**
 - For NMOS: $W_N/L = 0.375/0.25 = 1.5$
 - For PMOS: $W_P/L = 1.9375/0.25 = 7.75$ (where $W_P = 2.5 \times W_N$)
- **Significance:** Different W/L ratios are chosen to contrast SPICE simulation results and highlight how sizing affects inverter performance.



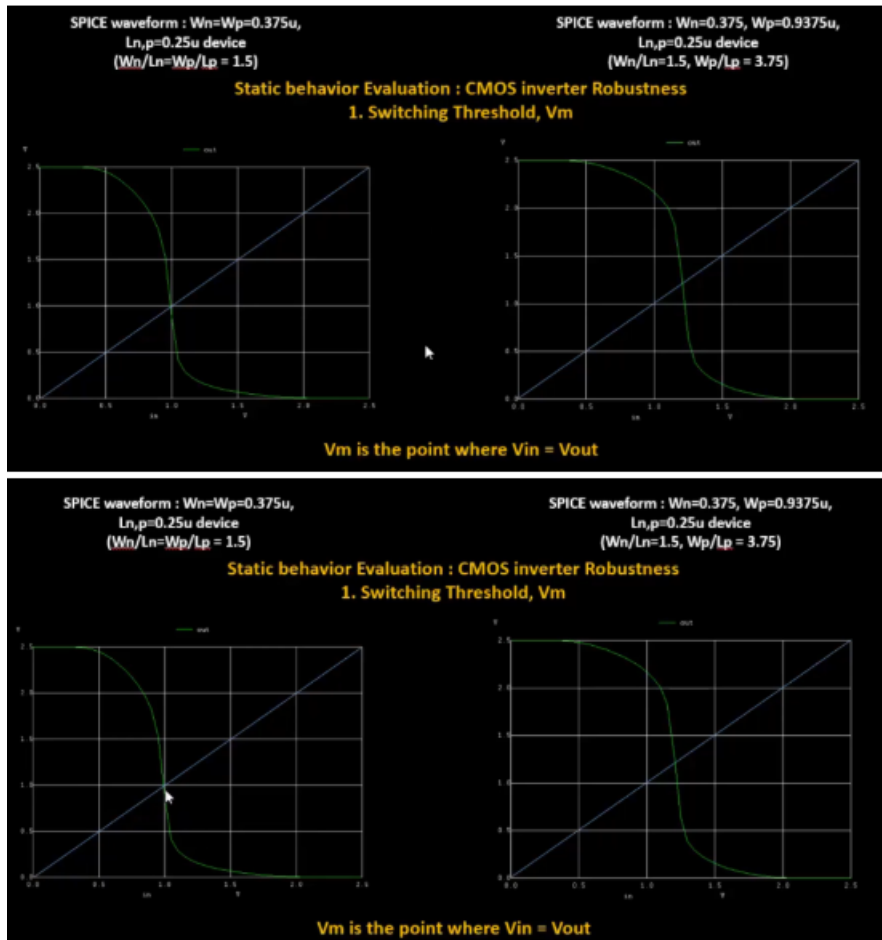
Robustness of CMOS Inverter

- **Waveform Similarity:** The qualitative shapes of output waveforms for both sizing cases are almost identical despite minor glitches. The essential switching behavior is preserved across different device sizes.
- **Key Feature:** CMOS inverters maintain fundamental logic—when input is 0, output is high, and when input is high, output is low—across a broad range of sizing configurations.
- **Application:** This intrinsic robustness is a principal reason CMOS is widely used for digital logic design.



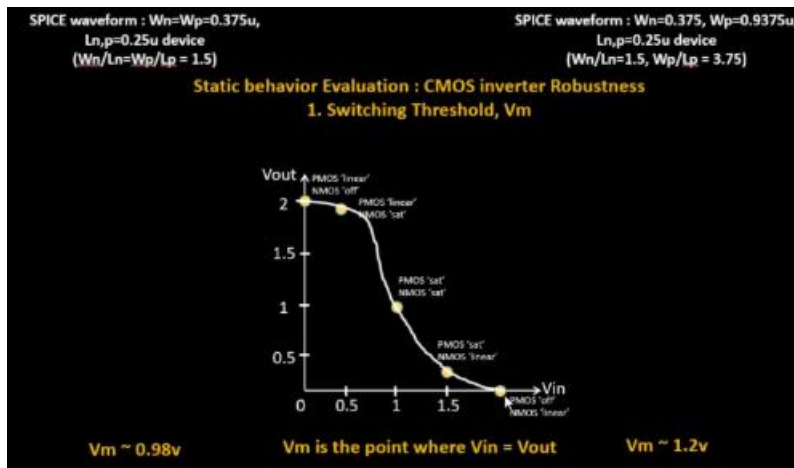
Switching Threshold

- **Definition:** The switching threshold is the input voltage (V_M) where the output voltage equals the input voltage ($V_{in} = V_{out}$).
- **Geometric Interpretation:** On the voltage transfer curve, the threshold is found at the intersection point where the output equals the input, typically visualized by drawing a 45° line ($V_{out} = V_{in}$).
- **Typical Observations:**
 - For equal W/L: Switching threshold found near 0.98 V.
 - For larger PMOS: Switching threshold shifts upward, near 1.2 V (not centered), reflecting the impact of width ratio on inverter symmetry.
- **Critical Region:** At the switching threshold, both PMOS and NMOS operate in the *saturation region* and are essentially turned on.



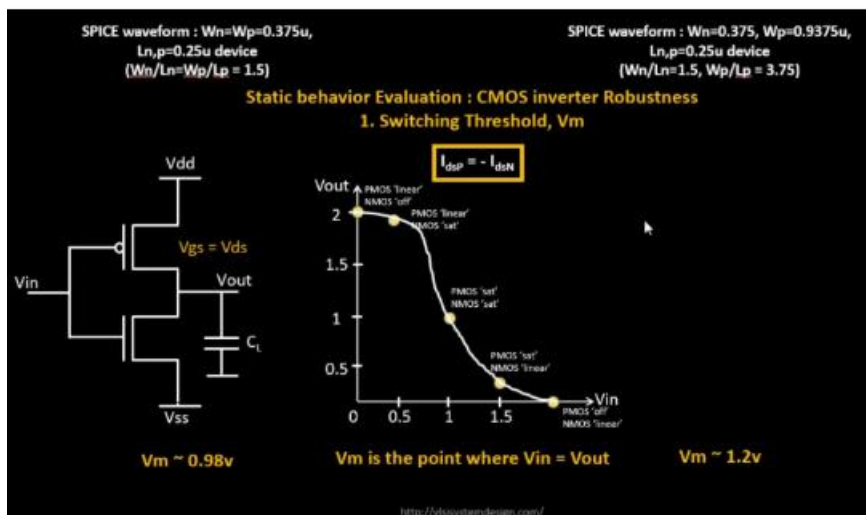
aturation Region and Current Flow

- **Saturation Region:** At threshold, the gate-source voltage (V_{GS}) substantially exceeds the threshold voltage, driving both transistors into saturation.
- **Current Behavior:**
 - Both transistors conduct, potentially causing substantial current from power to ground (short-circuit condition).
 - Drain currents have inverse directions:
 - PMOS: I_{DSP}
 - NMOS: I_{DSN}
 - At the threshold point: $I_{DSP} = -I_{DSN}$
- **Equations:**
 - Operating point condition: $V_{GS} \gg V_{TH}$
 - Threshold operation: $V_{GS} = V_{DS}$



Analytical Relationships

- **Threshold Calculation:**
 - Knowing W_N/L and W_P/L , the switching threshold V_M can be analytically derived.
 - Conversely, to set a desired V_M , appropriate values for device sizing can be chosen.
- **Physical Significance:** Control over threshold by sizing enables optimization for different applications (speed, power, noise margin).



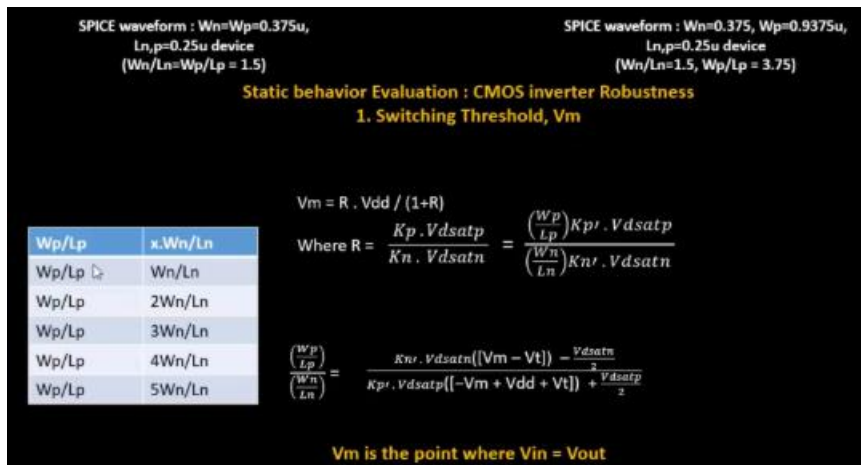
Key Points

- CMOS inverters remain robust across diverse sizing choices.
- Switching threshold (V_M) is a critical parameter determined by transistor sizing; it can be tuned via W/L ratios.
- At threshold, both PMOS and NMOS are in saturation, creating a region of potentially high current flow.

- The direction and magnitude of currents in PMOS and NMOS are opposite but equal at the switching point ($I_{DSP} = -I_{DSN}$).
- Threshold and robustness analysis are foundational for optimizing logic gate design in physical circuits.

Static and dynamic simulation of CMOS inverter

This section explores the concept of **switching threshold** in CMOS inverters, its robustness against variations in PMOS/NMOS sizing, and fundamental methods to analyze delay characteristics using theoretical and simulation-driven approaches. The importance of the switching threshold, the construction and interpretation of DC transfer characteristics, and the core methodology behind transient (dynamic) analysis are covered.



Core Concepts

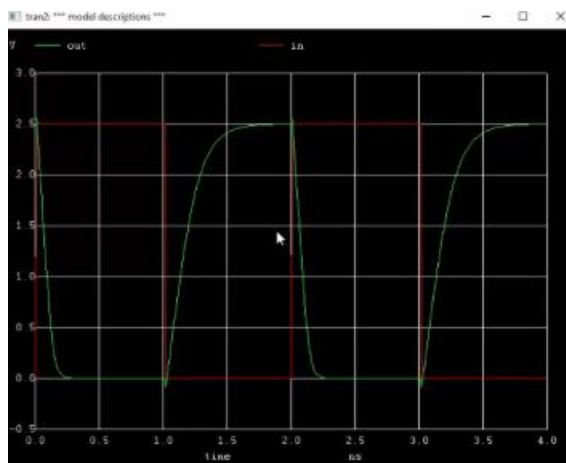
Switching Threshold in CMOS Inverters

- The **switching threshold** is the input voltage at which the output voltage of a CMOS inverter transitions between logic levels (high-to-low or low-to-high).
- At this threshold, the voltage at the input equals the voltage at the output: $V_{in} = V_{out}$.
- A common graphical technique to find the switching threshold is to plot the DC transfer characteristic and superimpose a 45° line (i.e., $V_{in} = V_{out}$). The intersection point gives the threshold voltage.
- The robustness of the threshold refers to how stable this value remains when PMOS or NMOS transistor dimensions are changed.

- In device terms, the **switching threshold** influences noise margins and affects the overall reliability of digital logic circuits.

DC Transfer Characteristics

- The **DC transfer characteristic** of a CMOS inverter plots output voltage (V_{out}) versus input voltage (V_{in}) under steady-state (DC) conditions.
- This curve typically shows a sharp transition from V_{DD} (logic high) to 0 V (logic low), centered around the switching threshold.
- The slope of this curve at the switching threshold is an important metric, as it relates to the inverter's noise immunity.
- Comparing how this characteristic shifts as PMOS and NMOS sizes are changed provides insight into inverter robustness and balancing.

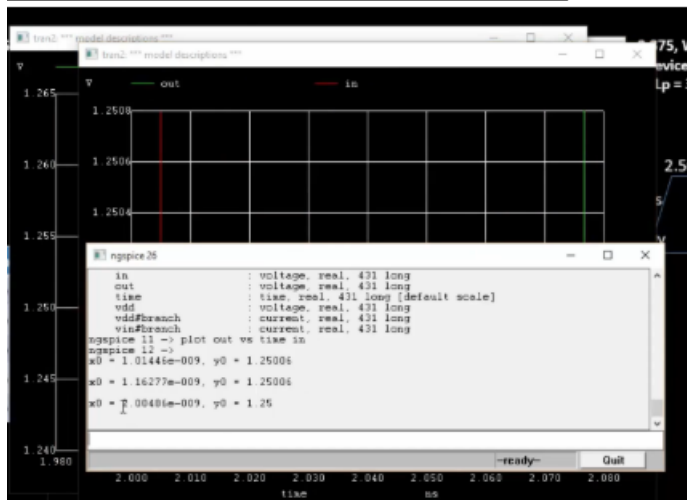
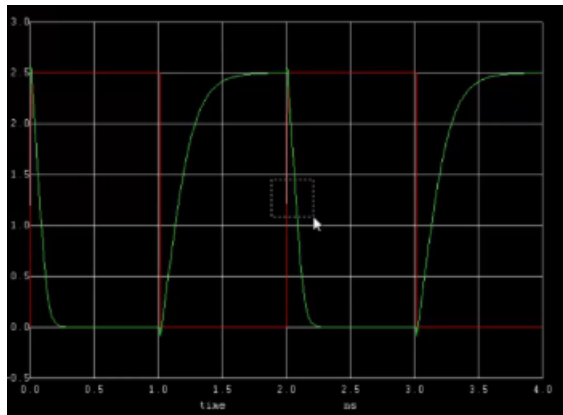


Delay in CMOS Inverters

- **Delay** in a CMOS inverter refers to the time taken for the output to respond to changes in the input.
- Two important types:
 - **Rise Delay (t_{PLH}):** Time taken for the output to rise from low to high (usually measured at the 50 points).
 - **Fall Delay (t_{PHL}):** Time taken for the output to fall from high to low (also at the 50 points).
- Delay is determined by the intrinsic device capacitance, the drive strength (dependent on PMOS/NMOS sizing), and the load being switched.
- Delay values can be extracted by noting the time difference between the input and output waveforms crossing the 50 voltage level in transient simulation.

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$$\$delay = t_{\{out,50\}} - t_{\{in,50\}}\$$$

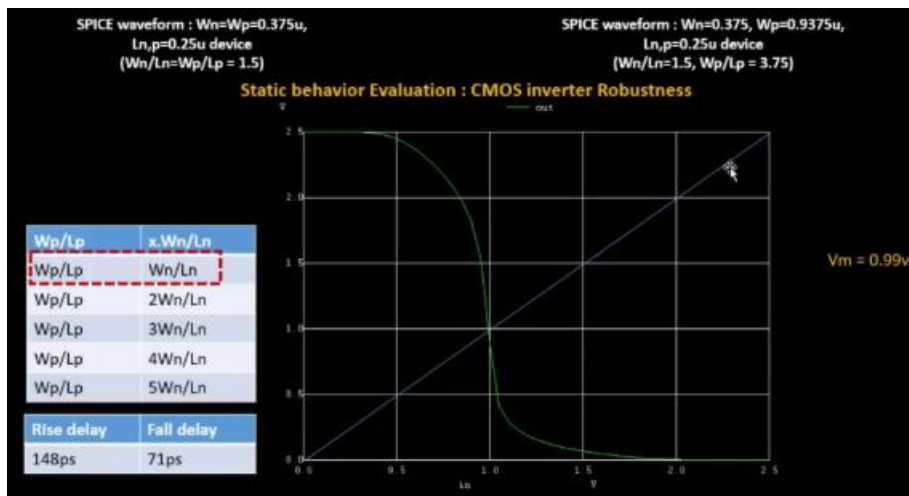


Transient (Dynamic) Analysis

- **Transient analysis** studies circuit response over time to dynamic input conditions, such as a pulse or step.
- In simulations, an input pulse (often with specified rise and fall times, width, and period) is applied, and the inverter's output is monitored.
- Key parameters for the input pulse:
 - **Rise time (t_{rise})**
 - **Fall time (t_{fall})**
 - **Pulse width**
 - **Period**

These parameters shape how quickly the input transitions and thus affect observed timing results.

- The transient simulation records output voltage over time, allowing measurement of **propagation delays**.



Key Points

- Switching Threshold:** The input voltage where $V_{in} = V_{out}$; found using the intersection of the DC transfer curve and the 45° line.
- Robustness Assessment:** By sweeping PMOS/NMOS widths, the switching threshold's stability gives insight into circuit robustness against sizing variations.
- Delay Measurement:** Propagation delays are computed as the difference in time stamps where input and output signals cross 50% of their logic swing.
- Dynamic Simulation:** Transient analysis enables direct visualization of timing behavior (rise/fall delays) in response to realistic input pulses.
- Importance for Design:** Both threshold and delay are central to noise margins, signal integrity, and overall digital performance in open-source hardware.